

# CMPDI DETAILED PROJECT REPORT (DPR) - Magadh Opencast CCL

Subsidiary: CCL | Capacity: 20.0 MT | Geological Basin: North Karanpura Coalfield, Jharkhand

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This Detailed Project Report (DPR) formulates the mine planning, equipment sizing, and geological stripping layout for Magadh Opencast CCL under CCL.

Geological Reserve & Setting: Situated in the North Karanpura Coalfield, Jharkhand, the mine exploits thick coal seams belonging to the Barakar Formation. Total geological reserves are estimated at 480 Million Tonnes.

Rated Production Capacity: 20.0 MT under mechanized Opencast operations. Stripping ratio is projected at 1.85 m3/t, utilizing 42 m3 electric rope shovels, 240T rear dump trucks, and high-capacity bucket wheel systems.

Environmental Mitigation & FMC: Coal evacuation is planned via a First Mile Connectivity (FMC) rapid loading silo with in-pit crushing and a 4.2 km overland conveyor to eliminate surface haul truck emissions.

| Technical Parameter    | Project Specification     | DGMS / Ministry Compliance       |
|------------------------|---------------------------|----------------------------------|
| Subsidiary / Mine Name | CCL - Magadh_Opencast_CCL | Approved by CIL Board            |
| Rated Annual Capacity  | 20.0 MT                   | Environmental Clearance Granted  |
| Mining Method          | Opencast                  | High-wall Stability Factor > 1.3 |
| Stripping Ratio        | 1.85 m3/t                 | Continuous Reclamation Mandated  |
| Evacuation Mode        | Rapid Loading Silo / FMC  | Zero Surface Dust Spillage       |